

QuMOS: Quantum Computing for Accelerated Multi-Omics Analysis in Vaccine Development

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Abstract—Multi-omics datasets that integrate genomics, transcriptomics, and proteomics are critical for identifying viral mutations and guiding vaccine design, yet their size and complexity strain classical machine learning methods. We present QuMOS, a quantum machine learning framework applied to the NCBI SARS-CoV-2 spike protein FASTA dataset. Amino acid sequences were pre-processed, binary encoded, and mapped to quantum circuits using a ZZFeatureMap, producing kernels for classification with quantum support vector machines (qSVM). In Qiskit simulation, qSVM preserved kernel separability of distinct spike protein variants and maintained average state fidelity above 0.9 across repeated trials. Grover’s algorithm was implemented for mutation marker retrieval, achieving quadratic query complexity $O(\sqrt{N})$ compared to classical $O(N)$. A baseline comparison with classical SVMs showed that although accuracy increased with dataset size, training time and feature expansion grew steeply. In contrast, quantum kernels operated efficiently in exponentially large Hilbert spaces without explicit computation. Fidelity testing with SWAP circuits confirmed that mutation-relevant features remained stable under entanglement. Finally, we estimate that naïve binary encoding of the full $\sim 1,200$ amino acid spike sequence would require thousands of qubits, motivating dimensionality reduction and subsequence encoding for near-term devices. These results establish QuMOS as a proof-of-concept that quantum-enhanced classification and search algorithms can handle realistic genomic datasets, offering a scalable complement to classical bioinformatics and accelerating analysis in vaccine-focused multi-omics research.

Index Terms—Quantum Computing, Multi-Omics, SARS-CoV-2 Spike Protein, Quantum Machine Learning, Grover’s Algorithm, qSVM

I. CLINICAL RELEVANCE

Genomic and proteomic analysis of the SARS-CoV-2 spike protein is central to vaccine design. By encoding spike protein datasets into quantum circuits, QuMOS demonstrates that quantum feature maps and Grover’s search can preserve fidelity in mutation detection while offering improved computational scalability. These approaches may ultimately shorten discovery cycles in vaccine development.

II. METHODS

Dataset and Preprocessing: The NCBI SARS-CoV-2 spike protein dataset was obtained in FASTA format. Amino acid sequences were converted into binary encodings and mapped to qubits. To reduce dimensionality, subsequences and mutation markers were prioritized over full-sequence expansion.

Quantum Encoding: A ZZFeatureMap was used to embed classical sequence features into quantum states. Quantum gates (Hadamard, Controlled-Z, SWAP) were implemented in Qiskit

to create superposition, entanglement, and fidelity-preserving state manipulation.

Classification: A quantum support vector machine (qSVM) was constructed using the encoded kernel to classify mutation variants.

Search: Grover’s algorithm was implemented for mutation marker retrieval, demonstrating quadratic speedup relative to linear classical search.

Evaluation: Fidelity was tested using SWAP circuits to validate that encoded states preserved genomic information. Qubit scaling requirements were estimated for full-sequence encoding.

III. RESULTS

QuMOS established that genomic spike protein sequences could be successfully encoded into a quantum feature space and classified via qSVM. In simulation, kernel separability was preserved, and average state fidelity remained above 0.9 across repeated runs. Grover’s algorithm produced quadratic speedup in simulated retrieval tasks. Feasibility analysis revealed that full binary encoding of $\sim 1,200$ amino acids would require thousands of qubits, underscoring the importance of dimensionality reduction and feature selection for near-term devices.

IV. DISCUSSION & CONCLUSION

QuMOS demonstrates a proof-of-concept for quantum machine learning applied to vaccine-relevant multi-omics datasets. Unlike classical kernels, which become computationally expensive at scale, quantum kernels and Grover’s search leverage superposition and entanglement to analyze mutation patterns efficiently. While current results are limited to simulation, these findings highlight how quantum computing can complement classical bioinformatics for early vaccine development. Future work will focus on optimized encodings, expanded datasets, and benchmarking qSVM performance against classical SVMs.

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